EPSR-test-ex13 Tethered molecules and dihedral (DIH) groups

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises two `.ato` features of the `epsr` engine [1,2]: tethered molecules, whose tether atom must stay within `tol` of a fixed point, and `DIH` rotational groups, whose rotations carry a dihedral-angle restraint. Users tether molecules to model adsorption sites or fixed solutes and use DIH groups to keep a conformation [3].

Code exercised

Fortran (original, fortran_src/):

- `ato_read_and_save.f`: `minitc` (`T00001` flag, `tol`, DIH groups), `moloutc` (writes them back), `normalise_coordinates` (wrapping of tether points)
- `molfit.f`: `check_tethered` (move rejection), `en_mole_dihedral`, `en_dihedral`, `molfit`
- `ato_manipulation_rev4.f`: `en_tethered`, `fmolec1` (`ftethered` reported in `.out`)

Rust (port, rust/epsr-core/src/):

- `ato.rs`: `State::minitc`, `moloutc`, `normalise_coordinates`
- `energy.rs`: `check_tethered`, `en_tethered`, `en_change_tethered`, `en_mole_dihedral`, `en_dihedral`
- `montecarlo.rs`: `molfit`; `epsr.rs`: `wrap_molecules`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. In this case atom 1 (S) of each of the first 20 DMSO molecules is tethered to the initial molecular centre with tolerance `tol` = 0.5 Å, and both methyl groups of all 200 DMSO molecules are `DIH` groups about the C-S axis with an O-S-C-H dihedral target of 180°, restrained by (cos phi - cos phi0)² + (sin phi - sin phi0)².

Inputs

Copy of EPSR-test with only the `.ato` changed: `tol` 0.5 (first value of header line 2), the 20 first DMSO lines get `T00001` and tether coordinates equal to their centres, and every `ROT` group is turned into a `DIH` group with one dihedral `4 5 180.000` / `4 8 180.000`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
Three chained passes, each re-reading the tether flags, tether points and DIH groups written by `moloutc`. In the Fortran reference the `.out` reports a non-zero tethered energy (7.0E-06 per molecule), the molecule centres moved while the tether points are unchanged, and the 20 `T00001` flags and 400 DIH groups are written back. All 27 output files are identical after every pass.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.